Auteur: Wout Vanderborght
Studierichting: Informatica
Oefeningenreeks 5D nr. 3.2

$$f(x) = \frac{1}{x},\ g(x) = \frac{1}{x^2} \text{ in } \left[\frac{1}{2}, 2\right]$$

$$S = \int\limits_{\frac{1}{2}}^{1} \left(\frac{1}{x^2} - \frac{1}{x}\right) + \int\limits_{1}^{2} \left(\frac{1}{x} - \frac{1}{x^2}\right)$$

$$= \int\limits_{\frac{1}{2}}^{1} \frac{dx}{x^2} - \int\limits_{\frac{1}{2}}^{1} \frac{dx}{x} + \int\limits_{1}^{2} \frac{dx}{x} - \int\limits_{1}^{2} \frac{dx}{x^2}$$

$$= \left[\frac{-1}{x}\right]_{\frac{1}{2}}^{1} - [\ln|x|]_{\frac{1}{2}}^{1} + [\ln|x|]_{1}^{2} - \left[\frac{-1}{x}\right]_{1}^{2}$$

$$= 1 + \ln\frac{1}{2} + \ln 2 - \frac{1}{2}$$

$$= \frac{1}{2}$$

Tekening:

`5D-3.2-wout.vanderborght.INF.png`

References